When constant g is not enough

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Recognize the limits of the near-surface mgh model.

SUCCESS CRITERIA - I can...

1. Compare altitude with planetary radius.
2. Explain that gravitational field weakens with center distance.
3. Choose a model appropriate to the height interval.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A cart begins with 15 J kinetic energy and gains 7 J net work. Find final kinetic energy.

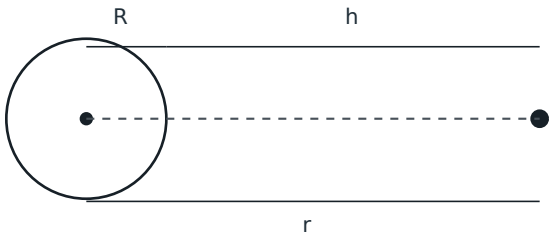
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta U \approx mg\,\Delta h$; $g = GM/r^2$.
The first relation needs nearly constant g. For large changes in r, the field changes; one fixed surface value is insufficient.

Worked reasoning: A local model treats g as constant over small height changes. At a center distance twice a planet's radius, however, g is only one quarter of its surface value.

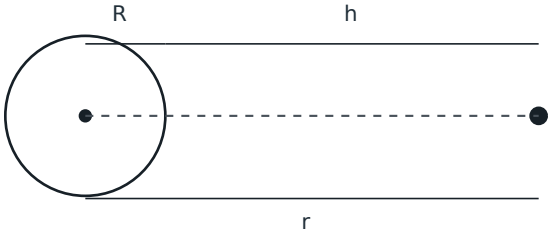
Your turn: Sketch a planet and mark center distances R and 2 R. Explain why one surface g cannot describe the entire trip exactly.



Marzano check - circle: ① ② ③

03 Read and interpret the model

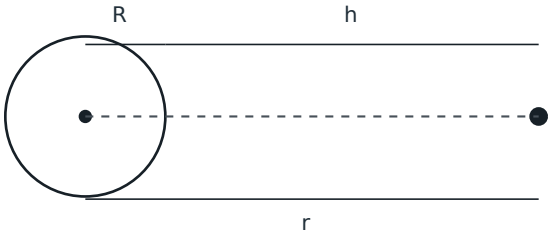
Compare lifting a book 1 m with moving a satellite one planetary radius above the surface. Which better fits constant g and why?



Marzano check - circle: ① ② ③

04 Apply the physics

Surface g is 12 N/kg. At center distance 2 R it is 3 N/kg. Would using 12 N/kg for the entire trip overestimate or underestimate required lifting energy? Explain. Label the relevant heights or distances on the model.



Marzano check - circle: ① ② ③

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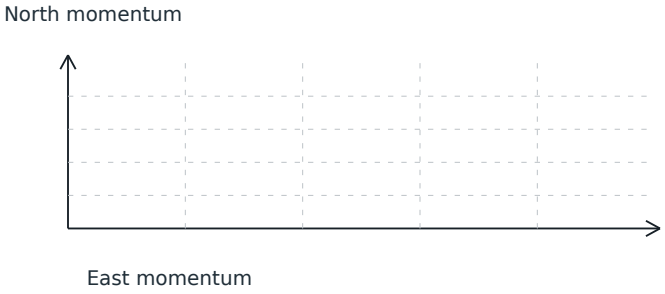
05 Explain your reasoning

Why is altitude h not interchangeable with center-to-center r in the universal expression?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

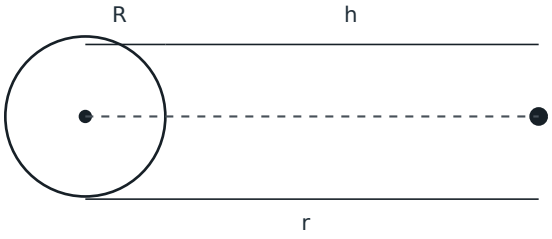
A system’s total momentum is 24 kg·m/s west and 32 kg·m/s south. Its combined mass is 4 kg. Find final velocity components.
Label momentum components and their sum with units.



Marzano check - circle: 1 2 3

07 Exit check

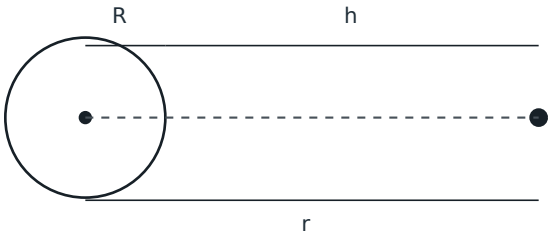
Choose the better description for a small classroom lift: constant g / strongly changing g. State a reason. Label the relevant heights or distances on the model.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A planet's surface field is 12 N/kg. Compare it at altitudes much smaller than R and at h=R. Which is better suited to constant-g reasoning? Label the relevant heights or distances on the model.



Marzano check - circle: 1 2 3